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Evaluating the efficacy of fungicides for the management of Sorghum (*Sorghum bicolor*) anthracnose (*Colletotrichum spp.*) in West Gondar

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Abstract

Sorghum is a critical food crop in Ethiopia, yet its productivity is severely threatened by anthracnose, caused by *Colletotrichum sublineolum*. The present study was conducted in West Gondar, Amhara Region, during the 2022 growing season, with field experiments established at Metema and West Armachiho. A randomized complete block design with three replications was employed to evaluate the efficacy and profitability of various fungicide regimes. Treatments included both single and combined applications of Thiram, Tilt, Redomil, and Natura, delivered as seed dressings, foliar sprays, or their combination, alongside an untreated control. The experiments utilized the susceptible local cultivar 'Tewzale' in 4×3 m plots, with 5 cm intra-row and 75 cm inter-row spacing. Statistical analysis indicated highly significant differences among treatments for key parameters: percent severity index (PSI), area under the disease progress curve (AUDPC), yield, and thousand seed weight (TSW) ($p < 0.0001$). Both location and treatment×location interactions were significant, especially for PSI at 65, 75, and 86 days after planting. The most effective disease suppression and lowest AUDPC values were observed with Tilt plus Thiram and Natura plus Thiram treatments, while untreated controls experienced the highest disease pressure. Notably, the Natura plus Thiram combination resulted in the highest grain yield (4614.6 kg ha⁻¹) and TSW (17.7 g), with the untreated control producing the lowest values (2076.6 kg ha⁻¹). Relative yield loss was completely mitigated by Natura plus Thiram, whereas the control plots recorded up to 61.8% loss. Economic assessment through partial budget analysis revealed that Natura plus Thiram was the most profitable option, delivering a net benefit of 79,162 ETB and a marginal rate of return of 2881.33%. Natura alone and Redomil plus Thiram also generated substantial returns, but to a lesser extent. In conclusion, integrating seed dressing with foliar fungicide applications specifically the Natura plus Thiram combination proved to be both an effective and economically viable strategy for managing sorghum anthracnose. For optimal results, this approach should be complemented by the use of resistant varieties and adherence to sound agronomic practices.

Keywords Sorghum, Anthracnose, *Colletotrichum spp.*, Fungicide efficacy, Seed dressing Foliar spray

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Introduction

Africa is the birthplace of sorghum (*Sorghum bicolor* L.), a cereal grain that is currently farmed in semi-arid tropical and temperate climates worldwide [18]. Sorghum accounts for over 65% of the carbohydrate requirements and 39% of the daily calorie intake for millions of people in developing countries [8]. Although sorghum is a staple grain for millions of people in Africa and India, the majority of sorghum utilized in the industrialized world is for livestock feed [18].

The need for sorghum products suggests that national sorghum improvement programs should boost and maintain high levels of sorghum production, particularly in Africa, where many populations are at risk. East and Central Africa (ECA) has very poor sorghum yields; productivity there is 1183 kg ha⁻¹, which is lower than yields in the United States (4354 kg ha⁻¹) [7].

The low productivity is in part caused by several biotic stresses especially, Striga, diseases and pests being the major challenges [24]. However, the diseases are exacerbated in the tropics of Africa by high rainfall and high relative humidity, moderate temperatures, and large amounts of inoculum [16].

When sorghum is grown in a warm, humid climate, sorghum anthracnose is common. In all of West Central Africa, it is extremely common and regarded as being of utmost importance [12]. Serious outbreaks are happening in farmers' and researchers' fields due to the foliar stage of the disease, which can affect many local landraces and improved cultivars [13, 23]. The northern Guinea and Sudan zones are where it is most common and destructive [17]. In West Africa, the disease results in yield losses that are both direct and indirect [22]. Depending on the cultivar, climate, and locality, the extent of direct losses varies. Depending on the cultivar, climate, and locality, the extent of direct losses varies.

The most significant factors reducing production are early seed abortion and a decrease in 1000-seed mass and density. In susceptible cultivars, foliar anthracnose-induced early leaf drying and defoliation can lower sorghum grain and fodder yields by 30–50%, or more, during severe epidemics [14]. In Burkina Faso, secondary infection of the stalk by the red rot phase may cause yield losses proportional to the severity of leaf anthracnose and stalk rot [15].

Indirect losses are also caused by grain anthracnose, which lowers seed germination and spreads the illness to other areas; however, the sub-yield region's loss from panicle anthracnose has not been identified [10]. Further, sorghum anthracnose often occurs as a mixed infection with zonate spot in Mali and Nigeria [11]. Seed treatment with Apron-plus (metalaxyl + carboxin + furathiocarp) alongside foliar-applied fungicides such as carbendazim + maneb and mancozeb has been reported

to be effective in the control of anthracnose in Nigeria [1]. Anthracnose in sorghum has also been successfully managed in experimental and seed production plots in Samaru, Nigeria, by applying two to three benomyl foliar sprays [9].

The best method for breeding sorghum foliar diseases is genetic resistance, particularly among farmers who provide resources [20]. However, resistance is hampered by the high genetic variability of pathogen populations [6]. There is very little information available on sorghum disease management techniques [20]. In Ethiopia, fungicide application during vegetation can greatly lower foliar diseases and aid in the development of grain yield and certain quality metrics [4]. Therefore, the objective of this trial is to evaluate the efficacy of fungicides for suppression of sorghum anthracnose disease.

Materials and methods

Description of test locations

The experiment was conducted during the 2022 main cropping season in the Metema and West Armachiho districts of the West Gondar Zone, Amhara Regional State, Ethiopia. The Metema experimental site is situated at 12°47'38" N and 36°23'41" E, at an elevation of 760 m above sea level. The area receives an average annual rainfall of 1030 mm, with maximum and minimum temperatures of 40.0°C and 15.0°C, respectively [2]. West Armachiho is located at 12°59'54" N and 36°10'57" E, with an elevation of 620 m above sea level. It has an average annual rainfall of 1109 mm, and maximum and temperatures of 38.2 °C and 19.3°C, respectively [3].

Treatments and experimental design

Four fungicides, administered as seed dressings, foliar sprays, or a combination thereof, along with an untreated control, were assessed in the hotspot regions of Metema and West Armachiho during the 2022 cropping season. Each fungicide was applied per the manufacturer's guidelines (Table 1). The experimental setup utilized a Randomized Complete Block Design (RCBD) with three replications. The local susceptible variety, Tewzale, was sown at a rate of 10 kg/ha, with a plot size of 12 m², 5 cm spacing between plants, and 75 cm between rows. NPS and urea were applied at rates of 121 kg/ha (fully at planting) and 100 kg/ha (in split applications) respectively. Disease severity was recorded by using a severity scale 1–9 (Table 2).

Data collected

Disease assessment

Severity The area or volume of plant tissue that is (visibly) diseased, usually relative to the total plant tissue, the

Table 1 Fungicides, mode of action and their application method used in the experiment

Treatments	Rate	Mode of action	Application method
1, Tilt plus Thiram	0.5Kg ha ⁻¹ and 3gm/kg	Systemic	Spray and Dressing
2, Tilt plus Undressed	0.5 L ha ⁻¹	Systemic	spray
3, Redomil plus Thiram	3 kg ha ⁻¹ and 3gm/kg	Systemic	Spray and dressing
4, Redomil plus Undressed	3 kg ha ⁻¹	Systemic	spray
5, Natura plus Thiram	0.5Lha ⁻¹ and 3gm/kg	Systemic	Spray and dressing
6, Natura plus Undressed	0.5 kg ha ⁻¹	Systemic	Spray
7, Thiram	3gm/kg		Dressing
8. unsparing and undressed	-	-	-

Table 2 Disease-scoring scales and reaction

Scale	Percentage (%)	Reaction
1	0	No Lesions or no hyper sensitive case
2	1—5	
3	6—10	
4	11—20	
5	21—30	Resistant
6	31—40	
7	41—50	Moderately Resistant
8	51—75	
9	> 75	Moderately susceptible
		Susceptible

Source: [21]

severity grades were converted into a percentage severity index (PSI) (DAP) for analysis.

AUDPC The area under disease progress curve (AUDPC) will be calculated for severity using Eq. 1:

$$\text{AUDPC} = \sum_{i=1}^{n-1} (0.5 (X_i + X_{i+1})) (t_{i+1} - t_i) \quad (1)$$

Where X_i is the severity index (PSI) of disease at i th assessment, t_i is the time of the i th assessment in days from the first assessment date and n is the total number of disease assessments (Campbell and Madden, 1990). Since incidence and severity will be expressed in percentage and time in days, (DAP), AUDPC will be expressed in %-days.

$$\text{PSI} = \frac{\text{Sum of individual numerical rating}}{\text{Total No. of plants assessed} \times \text{Max. score in scale}} \times 100$$

Yield and yield components

Grain Yield (GY) was measured from 3m² harvestable rows per plot using a sensitive balance, then adjusted to 12.5% moisture content with the formula incorporating actual moisture (measured by Dickey-John GAC 2100)

and standard moisture (12.5%). Thousand Seed Weight (TSW) was determined by weighing a random sample of 1,000 pure kernels using a sensitive balance.

Relative yield loss

Yield loss due to sorghum anthracnose was measured as percentage yield reduction of unsprayed plots compared with the most integrated plot using the following formula:

$$\text{RL (\%)} = \frac{(Y_1 - Y_2) \times 100}{Y_1}$$

Where, RL = relative loss (reduction of the parameters yield and yield components), Y_1 = mean of the respective parameter on maximum protected plot, Y_2 = mean of the respective parameter in other treatments and unprotected plot.

Cost and benefit assessment

Grain prices per kilogram and total sales from a hectare, fungicide prices, labor costs for spraying, and the cost of renting spray equipment were all taken into account. The local market was used to determine the grain price (Birr/kg), and the total sales from one hectare were calculated.

Partial budget analysis was used to do cost–benefit analysis based on the data. To determine the profitability of any new technologies that should be used in the agricultural industry, partial budgeting is used [5]. The process of choosing between different factor-product combinations while considering minor adjustments is known as marginal analysis. One metric that assesses how more capital spent under new management affects net returns in comparison to the prior one is the marginal rate of return. The marginal rate of return (MRR) was computed as follows to determine the increase in net return linked to each additional unit of cost (marginal cost)

$$\text{MRR} = \Delta \text{NI} / \Delta \text{IC}$$

Where, MRR is marginal rate of returns, ΔNI is change in net income compared with control, ΔIC is change in input cost compared with control.

Data analysis

Analysis of variance was done for disease parameters (severity for each days after planting (DAP), disease progress rate, area under disease progress curve) for each assessment date, yield and thousand seed weight. Least Significant Difference (LSD) values were used to separate differences among treatments means at 5% probability level. ANOVA was performed using General Linear Model (GLM) of SAS procedure [19]

Table 3 Disease severity and yield of treatment at Metema 2022

Treatments	DAP % 63	DAP %74	DAP % 84	AUDPC	TSW	Yield Kg ha ⁻¹
Tilt plus Thiram	28.7 ^{cd}	40.1 ^c	45.8 ^c	473 ^c	17.6 ^b	3774.1 ^{bc}
Tilt plus undressed	51.0 ^{ab}	49.2 ^{bc}	53 ^{abc}	562.1 ^{bc}	17.6 ^b	3660 ^{bc}
Redomil plus Thiram	29.4 ^{cd}	42 ^{bc}	57 ^{abc}	569.4 ^{bc}	17.8 ^{ab}	4386.5 ^{ab}
Redomil plus undressed	41.7 ^{bc}	48.6 ^{bc}	67.2 ^{abc}	636.9 ^{ab}	17.6 ^b	3863.5 ^{bc}
Natura plus Thiram	24.2 ^d	54.5 ^a	50.3 ^{abc}	544.5 ^{bc}	19.06 ^a	4691.9 ^a
Natura plus undressed	39.4 ^{bcd}	45.2 ^{abc}	58.2 ^{abc}	576.7 ^{bc}	17.8 ^{ab}	3339.7 ^c
Thiram	34.7 ^{bcd}	54.06 ^a	68.4 ^a	673.57 ^{ab}	14.9 ^c	2511.6 ^d
Untreated	58.13 ^a	64.6 ^a	74 ^a	762.67 ^a	13.2 ^d	2420.8 ^d
Mean	38.4	49.8	59.2	596.8	16.9	3581
LSD	16.37	12.8	21.9	148.4	1.4	790.6
CV	24.32	14.6	21.1	14.2	4.7	12.6

Note: the superscript letters (abc) indicate the same letter(s) are not significantly different at $P \leq 0.05$ according to the Least Significant Difference (LSD) test

Table 4 Disease severity and yield of treatment at West Armachiho 2022

Treatments	DAP%65	DAP% 75	DAP% 86	AUDPC	TSW	Yield Kg ha ⁻¹
Tilt plus Thiram	23.3 ^e	26.33 ^{cd}	21.4 ^f	580.87 ^d	15.4 ^a	3715.9 ^{bc}
Tilt plus undressed	38.93 ^b	54.6 ^a	49.3 ^c	990.12 ^b	15.9 ^a	3215.1 ^c
Redomil plus Thiram	35.6 ^{bc}	22.8 ^d	30.867 ^e	637.17 ^{dc}	14.8 ^a	3877.2 ^b
Redomil plus Undressed	25.53 ^{de}	26.2 ^{cd}	29.333 ^e	589.6 ^d	16.03 ^a	3570 ^{bc}
Natura plus Thiram	33.8 ^{bcd}	30 ^{bc}	24.83 ^{ef}	718.25 ^{dc}	16.3 ^a	4537.2 ^a
Natura plus undressed	27 ^{cde}	30.6 ^{bc}	37.4 ^d	698.67 ^{dc}	16.3 ^a	3986 ^{ab}
Thiram	40.86 ^b	33.6 ^b	60.6 ^b	820.43 ^{bc}	11.5 ^b	2572.9 ^d
Untreated	50.8 ^a	58.8 ^a	73.6 ^a	1226.2a	11.7 ^b	1732.3 ^e
Mean	34.4	35.3	40.91	782.66	14.7	3400.8
LSD	8.7292	6.1664	6.1251	204.27	1.5	612.37
CV	14.45	9.95	8.5	14.90	6.11	10.28

Values within a column followed by the same superscript letters (a, b, c, d, e, f) are not significantly different from each other according to the Least Significant Difference (LSD) test at $P \leq 0.05$.

Results and discussions

The Analysis of variance also shows of diseases severity in each day after planting and the area under diseases progress curve revealed that there were statistically significant ($P \leq 0.05$) difference among treatments both at Metema and at West Armachiho in the cropping season. The lowest diseases intensity (AUDPC) was observed on Tilt plus Thiram and Natura plus Thiram sprayed plots 473 and 544.5 respectively while the highest was recorded from unsprayed plot (762.7) in the cropping season at Metema (Table 3). Similarly, the same result was also found at West Armachiho on the disease intensity. The lowest diseases intensity (AUDPC) was observed on Tilt plus Thiram sprayed plots (580.9) while the highest was recorded from unsprayed plot (1226.2) in the cropping season (Table 4).

The analysis of variance of yield and disease intensities trait revealed that there were statistically significant at ($P \leq 0.05$) difference among treatments in yield and thousand seed weight in 2022 cropping season at Metema. The highest grain yield was recorded from Natura plus Thiram treated (4692 kg ha⁻¹) followed by Redomil plus Thiram sprayed (4386.5 kg ha⁻¹) while the lowest was recorded from unsprayed control (2420.8 kg ha⁻¹).

Similarly, the highest thousand seeds weight also found on Natura plus Thiram sprayed (19.1 g) treated treatment and followed by Redomil plus Thiram treated (17.8 g) while the lowest was recorded from unsprayed control (13.2) in the location (Table 3). Similarly, at West Armachiho the analysis of variance for yield and thousand seeds weight revealed that there were statistically significant ($P \leq 0.05$) difference among treatments. The highest grain yield was recorded from Natura plus Thiram treated (4537.2 kg ha⁻¹) followed by only Natura sprayed (39865 kg ha⁻¹) and Redomil plus Thiram sprayed (3877.2 kg ha⁻¹) while the lowest was recorded from unsprayed control (1732.3 kg ha⁻¹). In thousand seeds weight, highest gram was also found on Natura plus Thiram treated treatment (Table 4).

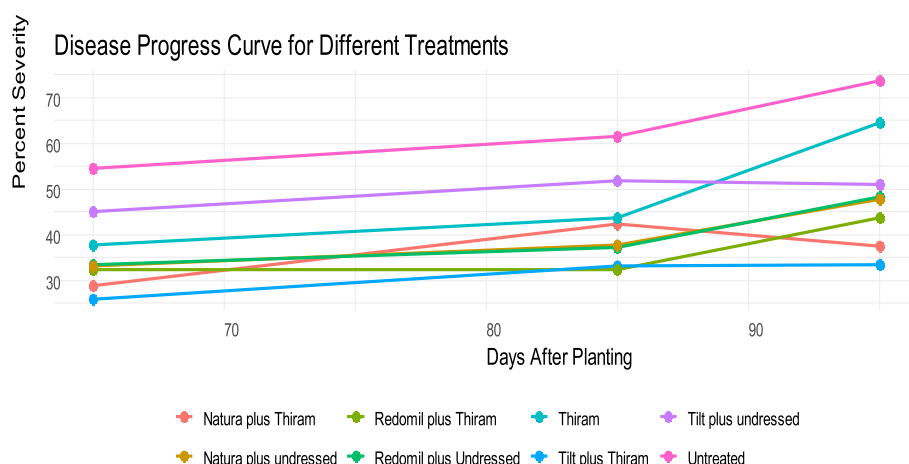
Combined analysis

The combined analysis of variance across locations revealed highly significant differences ($p < 0.0001$) among treatments for percent severity index (PSI), area under the disease progress curve (AUDPC), yield, and thousand seed weight (TSW), demonstrating clear variation in treatment efficacy. Location also had a highly significant impact on all disease parameters and TSW, while yield

Table 5 Combined disease intensities and yield of treatments at Metema and West Armachiho in 2022

Treatments	PSI 65(%)	PSI 75(%)	PSI 86(%)	AUDPC	TSW	YIELD (Kg ha ⁻¹)
Tilt plus Thiram	26.0 ^d	33.2 ^d	33.6 ^d	526.93 ^d	16.53 ^b	3745 ^{bc}
Tilt plus undressed	45 ^b	51.9 ^b	51.2 ^b	776.11 ^b	16.76 ^{ab}	3437.6 ^c
Redomil plus Thiram	32.5 ^{cd}	32.4 ^d	43.9 ^{bcd}	590.83 ^d	16.35 ^b	4131.9 ^b
Redomil plus Undressed	33.6 ^{cd}	37.4 ^{cd}	48.3 ^{bc}	613.25 ^d	16.81 ^{ab}	3716.7 ^{bc}
Natura plus Thiram	29.0 ^{cd}	42.3 ^c	37.6 ^{cd}	631.31 ^{cd}	17.7 ^a	4614.6 ^a
Natura plus undressed	33.2 ^{cd}	37.9 ^{cd}	47.8 ^{bc}	637.58 ^{cd}	17.08 ^{ab}	3662.8 ^c
Thiram	37.8 ^{bc}	43.8 ^{bc}	64.5 ^a	747 ^{bc}	13.23 ^c	2542.3 ^d
Untreated	54.5 ^a	61.7 ^a	73.8 ^a	994.43 ^a	12.45 ^c	2076.6 ^d
Mean	36.45	42.58	50.08	689.68	15.86	3490.921
LSD	8.8	8.29	10.9	121.38	0.98	465.77
CV	20.52	16.52	18.46	14.9	5.24	11.31
Trt	<.0001	<.0001	<.0001	<.0001	<.0001	<.0001
Trt*loc	0.0273	0.0032	0.0178	0.0023	0.1580	0.3246

PSI percent severity index, AUDPC Area under disease progress curve, TSW Thousand seed weight, Trt treatments, and Trt*Lo = treatment by location

**Fig. 1** Disease progress curve

differences across locations were significant (Table 5), confirming environmental diversity.

The treatment × location interaction was highly significant for PSI at 65, 75, and 86 days after planting ($p < 0.0001$), indicating that treatment effectiveness was influenced by environmental conditions. At 65 DAP, the lowest PSI values were recorded in Tilt plus Thiram (26.0%) and Natura + Thiram (29.0%). At 75 DAP, Redomil plus Thiram (32.4%) and Tilt plus Thiram (33.2%) showed the lowest PSI, while at 86 DAP, the lowest severity was again observed in Tilt plus Thiram (33.6%) and Natura plus Thiram (37.6%). The untreated control consistently showed the highest disease severity at all-time points.

Regarding AUDPC, Tilt plus Thiram (526.93) and Redomil plus Thiram (590.83) recorded the lowest values, reflecting superior disease suppression. Yield was highest in Natura + Thiram (4614.6 kg ha⁻¹) and Redomil plus Thiram (4131.9 kg ha⁻¹), while the untreated control had the lowest yield (2076.6 kg ha⁻¹). Similarly, TSW peaked in Natura plus Thiram (17.7 g) and was lowest in the untreated control (12.45 g). These findings underscore

the consistent benefits of combining fungicide sprays with seed treatment to effectively manage disease and enhance crop productivity.

As indicating disease progress curve Tilt plus Thiram consistently shows the lowest PSI values at all-time points. It had the slowest disease progression and the lowest average severity (30.93%), indicating it was most effective at controlling the disease (Fig. 1).

For each fungicide (Tilt, Redomil, Natura), combining with Thiram seed dressing reduced disease severity compared to the same fungicide applied to undressed seeds.

Relative yield loss

Yield loss due to Anthracnose was measured as percentage yield reduction of all other treatments plots compared with the most effective plot Natura plus Thiram. As compared to this reference, the maximum yield loss was obtained from untreated plots (22.7%) at Metema and West Armachiho 28.1% while the minimum was obtained from plots of only Natura sprayed (5.5%) at

Table 6 Relative yield loss of treatments by Anthracnose at Metema and West Armachiho in 2022

Treatments	RYL (%) at West Armachiho	RYL (%) at Metema	
Tilt plus Thiram	18.1	19.55	
Tilt plus undressed	29.18	21.98	
Redomil plus Thiram	14.55	6.51	Very good control, especially at Metema
Redomil plus undressed	21.3	17.65	
Natura plus Thiram	0	0	The most effective treatment overall
Natura plus undressed	12.15	28.8	
Thiram	43.32	46.46	
Untreated	61.81	48.41	Highest yield losses

RYL Relative yield Loss

Table 7 Partial budget analysis

Treatments	Tilt + Thiram	Tilt	Redomil + Thiram	Redomil	Natura + Thiram	Natura	Thiram	Control
Ajyield k/ha	3370.5	3093.8	3718.7	3345	4153.1	3296.5	2288.1	1868.9
Total gross benefit	67410	61876	74374	66900	83062	65930	45762	37378
Chemicals	2675	1675	2600	1600	2200	1200	1000	0
Dressing + spraying	700	500	700	500	700	500	200	0
Total variable cost	3375	2175	3300	2100	3900	1700	1200	0
Net benefit	64035	59701	71074	64800	79162	64230	44562	33867
MRR	0	0	1010.933	142.5	2881.333	3933.6	891.25	

The currency used is Ethiopian Birr, 1 kg sorghum is 20 Birr, MRR Marginal rate of return, Ajyield Kg ha⁻¹ adjusted yield

West Armachiho and 3.1% from Redomil plus and Thiram treated at Metema (Table 6).

Cost benefit analysis

The partial budget analysis indicated that the highest yield of Natura + Thiram was 4153.1 kg/ha, the highest net benefit was 79,162 ETB, and the MRR was 2881.33%, which is an excellent return on investment. The least effective performance Take charge The lowest yield, 1868.9 kg/ha, and the lowest net benefit, 33,867 ETB Cost-free, but yields a much lower return than treatments.

Chemical effectiveness while not as high as Natura-based combinations, Tilt plus Thiram and Redomil plus Thiram both produced positive MRR and good net benefits (64,035 and 71,074 ETB). Thiram's net benefit alone was 44,562 ETB, which was significantly less than combinations but better than control.

Priority in the economy economically, treatments with high MRR and net benefit are better. The best performer in terms of both biology and economics is Natura plus Thiram, Second Natura alone provided a respectable net benefit and a robust MRR (3933.6%), Thiram and Natura integration provides the maximum yield and profit, with a very high return on investment.

In addition, will have a Based on the dominant analysis, the dominant treatments were rejected from further economic analysis [5]. Considering the net benefit and the minimum acceptable MRR value (> 100%). Thus the stated treatment (application of Natura plus Thiram), will have the best economically

profitable practice for sorghum anthracnose disease management (Table 7).

Study limitations

Since this trial was only run during one cropping season, it might not accurately represent changes in crop performance, pathogen pressure, or weather from year to year. Because Colletotrichum sublineola population variability was not evaluated in this study, strain-specific variations in fungicide response are still unknown. The economic analysis was predicated on current grain prices and input costs, which could alter in subsequent growing seasons. Sustainability may be impacted by the failure to assess the long-term safety, residue levels, and environmental effects of repeated fungicide use.

Conclusion and recommendation

A combined ANOVA across locations confirmed distinct differences in efficacy by revealing highly significant treatment effects on yield, thousand seed weight, and anthracnose severity (PSI, AUDPC). For PSI, there was a significant interaction between location and treatment, suggesting that the environment affects performance. With an outstanding marginal rate of return (2881%), Natura + Thiram continuously produced the highest yield (4614.6 kg ha⁻¹), the highest thousand seed weight (17.7 g), the lowest disease severity, and the highest net benefit (79,162 ETB) among the treatments. Positive economic returns and effective disease suppression were also offered by

Tilt + Thiram and Redomil + Thiram. Conversely, the untreated control exhibited the highest yield loss (48–62%), lowest yield (2076.6 kg ha⁻¹), and the highest disease severity.

The better protection of Thiram-integrated treatments was validated by relative yield loss analysis, with Natura + Thiram preventing losses.

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Not applicable.

Authors' contributions

Author contributions: Misganaw Gelaye: Conceptualized the study, developed the research plan, performed data analysis, and drafted the initial manuscript. Asfaw Azanaw: Supervised the research, validated methodologies, and provided overall guidance and oversight. Yohannes Kefale: Conducted field research and data collection. Mintesnot Worku: Managed and organized the research data.

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Data availability

The datasets generated and/or analyzed during the current study are not publicly available due to privacy and confidentiality considerations. However, data may be provided by the corresponding author upon reasonable request, subject to the approval of a formal data-sharing agreement and compliance with ethical and regulatory requirements set by ARARI and Gondar Agricultural Research Center.

Declarations

Ethics approval and consent to participate

This study was reviewed and approved by the Ethics Committee of the Amhara Agriculture Research Institute (ARARI), Gondar Agricultural Research Center, written informed consent was obtained from all participants prior to their involvement in the study. All research activities were carried out in accordance with ethical standards outlined in the Memorandum of Understanding.

Consent for publication

All authors have read and approved the final manuscript, they have agreed to the submission and potential publication of this work and declare no disputes regarding authorship.

Competing interests

The authors declare no competing interests.

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